Lista 2 - Eletromagnetismo

1-a) $\frac{d\hat{a}_n}{d\phi} = a_\phi$ para coordenadas cilíndricas $\begin{cases} \hat{a}_n = \cos\phi\, \hat{a}_x + \operatorname{sen}\phi\, \hat{a}_y \\ \hat{a}_\phi = -\operatorname{sen}\phi\, \hat{a}_x + \cos\phi\, \hat{a}_y \\ A_z = A\hat{a}_z \end{cases}$

$$\frac{\hat{a}_n}{d\phi} = \frac{(\cos\phi + \operatorname{sen}\phi)}{d\phi} = -\operatorname{sen}\phi + \cos\phi = a_\phi$$

b) $\frac{d\hat{a}_\phi}{d\hat{a}_\phi} = -\hat{a}_n$

$\hat{a}_n = \cos\phi + \operatorname{sen}\phi$

$-\hat{a}_n = -\cos\phi - \operatorname{sen}\phi$

$$\frac{d\hat{a}_\phi}{d\hat{a}_\phi} = \frac{(-\operatorname{sen}\phi + \cos\phi)}{d\phi} = -\cos\phi - \operatorname{sen}\phi = -\hat{a}_n$$

c) $\frac{\partial \hat{a}_R}{\partial\theta} = \hat{a}_\theta$ coordenadas esféricas $\begin{cases} \hat{a}_R = \operatorname{sen}\theta\cos\phi + \operatorname{sen}\theta\operatorname{sen}\phi + \cos\theta \\ \hat{a}_\theta = \cos\theta\cos\phi + \cos\theta\operatorname{sen}\phi - \operatorname{sen}\theta \\ \hat{a}_\phi = -\operatorname{sen}\phi + \cos\phi \end{cases}$

$$\frac{\partial \hat{a}_R}{\partial\theta} = \frac{1}{\partial\theta}(\operatorname{sen}\theta\cos\phi + \operatorname{sen}\theta\operatorname{sen}\phi + \cos\theta) = \cos\theta\cos\phi + \cos\theta\operatorname{sen}\phi - \operatorname{sen}\theta$$

$$\hat{a}_\theta = \cos\theta\cos\phi + \cos\theta\operatorname{sen}\phi - \operatorname{sen}\theta$$

d) $\frac{\partial \hat{a}_\theta}{\partial\theta} = -\hat{a}_R$

$a_R = \operatorname{sen}\theta\cos\phi + \operatorname{sen}\theta\operatorname{sen}\phi + \cos\theta$

$-a_R = -\operatorname{sen}\theta\cos\phi - \operatorname{sen}\theta\operatorname{sen}\phi - \cos\theta$

$$\frac{d\hat{a}_\theta}{d\theta} = \frac{\cos\theta\cos\phi + \cos\theta\operatorname{sen}\phi - \operatorname{sen}\theta}{d\theta}$$

$$= -\operatorname{sen}\theta\cos\phi - \operatorname{sen}\theta\operatorname{sen}\phi - \cos\theta = -\hat{a}_R$$

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e) $\frac{\partial \hat{a}_R}{\partial \phi} = \sin\theta\, a_\phi$

$$\frac{\hat{a}R}{\partial \phi} = \frac{\sin\theta\cos\phi + \sin\theta\sin\phi + \cos\theta}{\partial\phi}$$

$$\hat{a}_R = -\sin\theta\sin\phi + \sin\theta\cos\phi$$

$$\hat{a}_R = \sin(\theta)(-\sin\theta\sin\phi + \sin\theta\cos\phi)$$

$$a_R = -\cancel{\sin^2\theta}\sin\phi + \cancel{\sin^2\theta}\cos\phi$$

$$a_R = -\sin\phi + \cos\phi = a_\phi$$

f) $\frac{\hat{a}_\theta \partial}{\phi \partial} = \cos\theta\, a_\phi$

$$\frac{\partial a_\theta}{\partial\phi} = \frac{\cos\theta\cos\phi + \cos\theta\sin\phi - \sin\theta}{\partial\phi}$$

$$\hat{a}_\theta = (-\cos\theta\sin\phi + \cos\theta\cos\phi) \equiv \cos\theta\, a_\phi$$

$$a_\theta = \cos\theta(-\cos\theta\sin\phi + \cos\theta\cos\phi)$$

$$= -\cancel{\cos^2\theta}\sin\phi + \cancel{\cos^2\theta}\cos\phi$$

$$= -\sin\phi + \cos\phi = a_\phi$$

2-a) $\vec{\nabla}\cdot(f\vec{A}) = A\cdot\vec{\nabla}f + f\vec{\nabla}\cdot\vec{A}$, $\vec{A} = A_x + A_y + A_z$

$f = f(x, y, z)$

$\vec{\nabla} = \left(\frac{\partial}{\partial x}\hat{a}x + \frac{\partial}{\partial y}\hat{a}y + \frac{\partial}{\partial z}\hat{a}z\right)$

$$\vec{\nabla}\cdot(f\vec{A}) = \vec{\nabla}\cdot(fA_x\hat{a}x + fA_y\hat{a}y + fA_z\hat{a}z)$$

$$\vec{\nabla}\cdot(f\vec{A}) = \left(\frac{fA_x}{\partial x}ax + \frac{fA_y}{\partial y}\hat{a}y + \frac{fA_z}{\partial z}\hat{a}z\right)$$

$$A\cdot\vec{\nabla}f = A\cdot\left(\frac{f}{\partial x} + \frac{f}{\partial y} + \frac{f}{\partial z}\right) \Rightarrow \left(\frac{f}{\partial x}\cdot A_x a_x + \frac{f}{\partial y}A_y\hat{a}y + \frac{f}{\partial z}A_z\hat{a}z\right)$$

$$f\vec{\nabla}\cdot A = f\cdot\left(\frac{A_x}{\partial x} + \frac{A_y}{\partial y} + \frac{A_z}{\partial z}\right) \Rightarrow \left(f\cdot\frac{A_x}{\partial x}\hat{a}x + f\frac{A_y}{\partial y}\hat{a}y + f\frac{A_z}{\partial z}\hat{a}z\right)$$

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$$\nabla\cdot(f\vec{A}) = \left(\frac{f A_x}{\partial x}\hat{a}_x + \frac{f A_y}{\partial y}\hat{a}_y + \frac{f A_z}{\partial z}\hat{a}_z\right)$$

$$A\cdot\vec{\nabla} f + f\vec{\nabla}\cdot\vec{A} = \left(\frac{f}{dx}A_x a_x + \frac{A_x}{\partial x} f\hat{a}_x + \frac{f}{\partial y}A_y\hat{a}_y + \frac{A}{\partial y} f\hat{a}_y + \frac{f}{\partial z}A_z\hat{a}_z + \frac{A}{\partial z} f\hat{a}_z\right)$$

$$= \left(\frac{f A_x}{\partial x}\hat{a}_x + \frac{f A_y}{\partial y}\hat{a}_y + \frac{f A_z}{\partial z}\hat{a}_z\right)$$

isto é possível graças as regras do calculo vetorial.

b) $\vec{\nabla}\times(f\vec{A}) = -A\times\vec{\nabla} f + f\vec{\nabla}\times\vec{A}$

$$\vec{\nabla}\times(f\vec{A}) = \vec{\nabla}\times(fA_x + fA_y + fA_z) = \left(\frac{\partial}{\partial x} + \frac{\partial}{\partial y} + \frac{\partial}{\partial z}\right)\times(fA_x + fA_y + fA_z)$$

$$\left(\frac{f A_z}{\partial y} - \frac{f A_y}{\partial z}\right)a_x + \left(\frac{f A_x}{\partial z} - \frac{f A_z}{\partial x}\right)\hat{a}_y + \left(\frac{f A_y}{\partial x} - \frac{f A_x}{\partial y}\right)a_z$$

$$-A\times\vec{\nabla} f = -(A_x + A_y + A_z)\times\left(\frac{f}{\partial x} + \frac{f}{\partial y} + \frac{f}{\partial z}\right)$$

$$= \left(-A_y\cdot\frac{f}{\partial z} + A_z\frac{f}{\partial y}\right)\hat{a}_x + \left(-A_z\frac{f}{\partial x} + A_x\frac{f}{\partial z}\right)\hat{a}_y + \left(-A_x\frac{f}{\partial y} + A_y\cdot\frac{f}{\partial x}\right)\hat{a}_z$$

$$f\vec{\nabla}\times A = \left(\frac{\partial}{\partial x} + \frac{\partial}{\partial y} + \frac{\partial}{\partial z}\right)\times(A_x + A_y + A_z)$$

$$= \left(f\frac{A_z}{\partial y} - f\cdot\frac{A_y}{\partial z}\right)\hat{a}_x + \left(f\cdot\frac{A_x}{\partial z} - f\cdot\frac{A_z}{\partial x}\right)\hat{a}_y + \left(f\cdot\frac{A_y}{\partial x} - f\cdot\frac{A_x}{\partial y}\right)\hat{a}_z$$

$$-A\times\vec{\nabla} F + f\vec{\nabla}\times A = \left(-A_y\frac{f}{dz} - \frac{A_y}{\partial z}\cdot f + A_z\frac{f}{\partial y} + \frac{A_z}{dy}f\right)\hat{a}_x + \left(-A_z\frac{f}{dx} - \frac{A_z}{dx}f + A_x\frac{f}{dz} + \frac{A_x}{dz}f\right)\hat{a}_y$$

$$+\left(-A_x\frac{f}{dy} - \frac{A_x}{dy}f + A_y\frac{f}{dx} + \frac{A_y}{\partial x}f\right)\hat{a}_z$$

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$$-A \times \vec{\nabla} F + F \vec{\nabla} \times A = \left(\frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z}\right)\hat{a}_x + \left(\frac{\partial A_x}{\partial z} - \frac{\partial A_z}{\partial x}\right)\hat{a}_y + \left(\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y}\right)\hat{a}_z$$

logo podemos concluir que $\vec{\nabla} \times (F\vec{A}) = -A \times \vec{\nabla} F + F \vec{\nabla} \times \vec{A}$

3- $\vec{F} = \vec{\nabla} F \Rightarrow \vec{F} = \left(\frac{\partial F}{\partial x}\hat{a}_x + \frac{\partial F}{\partial y}\hat{a}_y + \frac{\partial F}{\partial z}\hat{a}_z\right)$; $\vec{\nabla} = \left(\frac{\partial}{\partial x} + \frac{\partial}{\partial y}\hat{a}_y + \frac{\partial}{\partial z}\hat{a}_z\right)$

$\vec{F} \times \vec{\nabla} = 0$

$$\vec{F} \times \vec{\nabla} = \begin{vmatrix} a_x & a_y & a_z \\ F_x & F_y & F_z \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \end{vmatrix} = \left(\frac{\partial F_y}{\partial z} - \frac{\partial F_z}{\partial y}\right) + \left(\frac{\partial F_z}{\partial x} - \frac{\partial F_x}{\partial z}\right) + \left(\frac{\partial F_x}{\partial y} - \frac{\partial F_y}{\partial x}\right)$$

$$= \left(\frac{\partial F}{\partial z \partial y} - \frac{\partial F}{\partial y \partial z}\right)\hat{a}_x + \left(\frac{\partial F}{\partial z \partial x} - \frac{\partial F}{\partial x \partial z}\right)\hat{a}_y + \left(\frac{\partial F}{\partial x \partial y} - \frac{\partial F}{\partial y \partial x}\right)\hat{a}_z$$

$$0\,\hat{a}_x + 0\,\hat{a}_y + 0\,\hat{a}_z = 0$$

4- $\vec{A} = \vec{\nabla} \times \vec{F} \Rightarrow \vec{A} = \left(\frac{\partial F_z}{\partial y} - \frac{\partial F_y}{\partial z}\right)\hat{a}_x + \left(\frac{\partial F_x}{\partial z} - \frac{\partial F_z}{\partial x}\right)\hat{a}_y + \left(\frac{\partial F_y}{\partial x} - \frac{\partial F_x}{\partial y}\right)\hat{a}_z$

$$\vec{\nabla} \cdot \vec{A} = \left(\frac{\partial}{\partial x} + \frac{\partial}{\partial y} + \frac{\partial}{\partial z}\right) \cdot \left(\frac{\partial F_z}{\partial y} - \frac{\partial F_y}{\partial z}\right) + \left(\frac{\partial F_x}{\partial z} - \frac{\partial F_z}{\partial x}\right) + \left(\frac{\partial F_y}{\partial x} - \frac{\partial F_x}{\partial y}\right)$$

$$\left(\frac{\partial F \hat{a}_z}{\partial y} \cdot \frac{\partial}{\partial x}\hat{a}_x - \frac{\partial F \hat{a}_y \cdot \hat{a}_x}{\partial y \partial x}\right)\hat{a}_x + \left(\frac{\partial F \hat{a}_x \cdot \hat{a}_y}{\partial z \partial y} - \frac{\partial F \hat{a}_z \cdot \hat{a}_y}{\partial x \partial y}\right)\hat{a}_y$$

$$+ \left(\frac{\partial F \hat{a}_y \cdot \hat{a}_z}{\partial x \partial z} - \frac{\partial F \hat{a}_x \hat{a}_z}{\partial x \partial z}\right)\hat{a}_z$$

Sabemos que o produto de vetores unitarios ortogonais é 0, exemplo: $\hat{a}_1 \cdot \hat{a}_2 = 0$, portanto o produto das parcelas.

do produto vetorial entre $\vec{\nabla} \times \vec{A} = 0$, visto que $\vec{A} = \vec{\nabla} \times \vec{F}$.

5- Determine $\int_C (\operatorname{sen}\phi\, \hat{a}_r + r\cos\phi\, \hat{a}_\phi + \tan\phi\, a_z) \cdot d\vec{l}$

$$\begin{cases} r = 3 \\ z = 0 \\ \phi = 0 \leq \phi < \frac{\pi}{2} \end{cases}$$

$d\vec{l} = dr\, \hat{a}_r + r\, d\phi\, \hat{a}_\phi + dz\, \hat{a}_z$

$d\vec{l} = 0$

$$\int_0^{\frac{\pi}{2}} (\operatorname{sen}\phi\, \hat{a}_r + r\cos\phi\, a_\phi + \cancelto{0}{\tan\phi\, a_z}) \cdot r\, d\phi$$

$$= \int_0^{\frac{\pi}{2}} 3\operatorname{sen}\phi + 9\cos\phi\, d\phi \Rightarrow -3\cos\phi + 9\operatorname{sen}\phi \Big]_0^{\frac{\pi}{2}}$$

$$= \left[-3\cos\left(\frac{\pi}{2}\right) + 9\operatorname{sen}\left(\frac{\pi}{2}\right) - (-3\cos(0) + 9\operatorname{sen}(0))\right]$$

$$\left[(0 + 9 + 3\ \ 0)\right] = \boxed{12}$$

6- $\vec{F} = xy^2\, \hat{a}_x + yz^2\, \hat{a}_y + 2xz\, \hat{a}_z$, pontos $\begin{cases} A = (0,0,0) \\ B = (1,2,3) \end{cases}$,

determine $\int_C \vec{F} \cdot d\vec{l}$

$P = B - A = (1,2,3) - (0,0,0)$

$P = (1,2,3)$

$\vec{r}(t) = (1,2,3)t + (0,0,0)$

$\vec{r}(t) = (t, 2t, 3t),\ 0 \leq t \leq 1$

$\vec{F}(x(t), y(t), z(t)) = 4t^3\, \hat{a}_x + 18t^3\, \hat{a}_y + 6t^2\, \hat{a}_z$

$$\int_0^1 (4t^3_{a_x} + 18t^3_{a_y} + 6t^2_{a_z}) \cdot (dt_{a_x} + 2dt_{a_y} + 3dt_{a_z})$$

$$= \int_0^1 4t^3 dt + 36t^3 dt + 18t^2 dt$$

$$= \int_0^1 40t^3 dt + \int_0^1 18t^2 dt$$

$$d\vec{l} = \begin{cases} dt \\ 2dt \\ 3dt \end{cases}$$

$$\int_0^1 40t^3\,dt + \int_0^1 18t^2\,dt$$

$$= \frac{40t^4}{4}\Big|_0^1 + \frac{18t^3}{3}\Big|_0^1 = \left(\frac{40(1)^4}{4} - 0\right) + \left(\frac{18(1)^3}{3} - 0\right) = 70 + 6 = 76$$

7 - Prove que $\int_V \vec{\nabla}\cdot(\vec{\nabla}\times A)\,dV = 0$

O teorema de Stokes estabelece que para um campo vetorial $\vec{F}$ e uma superfície Σ delimitada por uma curva fechada C é:

$$\oint_C \vec{F}\cdot d\vec{l} = \iint_\Sigma (\vec{\nabla}\times\vec{F})\,d\vec{\Sigma}$$

já o teorema de Gauss estabelece que para um campo vetorial $\vec{F}$ e uma região V delimitada por uma Superfície Σ:

$$\oint_\Sigma \vec{F}\cdot d\vec{\Sigma} = \iiint_V \vec{\nabla}\cdot\vec{F}\,dV$$

logo podemos estabelecer a seguinte relação:

$$\oint_\Sigma (\vec{\nabla}\times\vec{F})\,d\vec{\Sigma} = \iiint_V \vec{\nabla}\cdot(\vec{\nabla}\times\vec{F})\,dV$$

Pelo teorema fundamental do Calculo de Divergencia, sabemos que $\vec{\nabla}\cdot(\vec{\nabla}\times\vec{F}) = 0$ para qualquer vetorial F. portanto temos

$$\oint_\Sigma (\vec{\nabla}\times\vec{F})\,d\vec{\Sigma} = \iiint_V \vec{\nabla}\cdot(\vec{\nabla}\times\vec{F})\,dV = 0$$

8- $\vec{A} = x^2 \hat{a}_x + yz\,\hat{a}_y + xy\,\hat{a}_z$, determine o fluxo desse vetor através da superfície definida nas condições $\{z = 4; 0 \leq x \leq 3; 1 \leq y \leq 2\}$ pelo teorema de gauss temos:

$$\oint_S \vec{F} \cdot d\vec{S} = \int_V (\vec{\nabla} \cdot \vec{F})\, dV$$

$$\vec{A} \cdot \vec{\nabla} = \langle x^2 \hat{a}_x + yz\,\hat{a}_y + xy\,\hat{a}_z \rangle \cdot \left\langle \frac{\partial}{\partial x}\hat{a}_x + \frac{\partial}{\partial y}\hat{a}_y + \frac{\partial}{\partial z}\hat{a}_z \right\rangle$$

$$(\vec{A} \cdot \vec{\nabla}) = 2x + z$$

$$\int_V \vec{\nabla} \cdot \vec{A}\, dV = \int_1^2 \int_0^3 2x + 4\, dx\, dy = \int_1^2 \frac{2x^2}{2} + 4x \Big|_0^3 dy \Rightarrow$$

$$= \int_1^2 9 + 12\, dy = \int_1^2 27\, dy = \left[27x\right]_1^2 = 42 - 27 = 27$$

9- $\vec{A} = x^2 y\, \hat{a}_x$

superfície é um cubo unitário $x = y = z = 1$

$$\oint \vec{F} \cdot d\vec{S} = \int_V \vec{\nabla} \cdot \vec{F}\, dV$$

$$(A \cdot \vec{\nabla}) = \langle x^2 y\, \hat{a}_x + 0 + 0 \rangle \left\langle \frac{\partial}{\partial x} + \frac{\partial}{\partial y} + \frac{\partial}{\partial z} \right\rangle = 2xy$$

$$\int_V \vec{\nabla} \cdot \vec{F}\, dV = \int_0^1 \int_0^1 \int_0^1 2xy\, dx\, dy\, dz = \int_0^1 \int_0^1 \left[\frac{2x^2 y}{2}\right]_0^1 = \int_0^1 \int_0^1 y\, dy\, dz$$

$$= \int_0^1 \left[\frac{y^2}{2}\right]_0^1 dz \Rightarrow \int_0^1 \frac{1}{2}\, dz = \frac{1}{2}\left[1 - 0\right] = \boxed{\frac{1}{2}}$$

10- $\vec{F} = R \operatorname{sen}\theta \, \hat{a}_R + 0 \, a_\theta + a_\phi$

$$\vec{\nabla} \cdot \vec{F} = \left\langle \frac{1}{R^2} \cdot \frac{\partial}{\partial R}\left(R^2 \cdot F_R\right) \hat{a}_R + \frac{1}{R \operatorname{sen}\theta} \cdot \frac{\partial}{\partial \theta}\left(\operatorname{sen}\theta \, F_\theta\right) \hat{a}_\theta + \frac{1}{R \operatorname{sen}\theta} \frac{\partial F_\phi}{\partial \phi} \right\rangle$$

$$= \left\langle \frac{1}{R^2} \cdot \frac{\partial}{\partial R}\left(R^3 \operatorname{sen}\theta\right) + \cancel{\frac{1}{R \operatorname{sen}\theta} \cdot \frac{\partial}{\partial \theta}\left(\operatorname{sen}\theta \cdot 0\right)}^{\,0} + \cancel{\frac{1}{R \operatorname{sen}\theta} \cdot \frac{\partial 0}{\partial \phi}}^{\,0} \right\rangle$$

$$= \frac{\operatorname{sen}\theta}{R^2} \cdot \frac{d}{\partial R}\left(R^3\right) \Rightarrow \frac{3R^{\cancel{2}} \operatorname{sen}\theta}{\cancel{R^2}} = \boxed{3 \operatorname{sen}\theta}$$

b) $\vec{\nabla} \times \vec{F}$

$$\det = \begin{vmatrix} \hat{a}_x & \hat{a}_y & \hat{a}_z \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ F_{ax} & F_{ay} & F_{az} \end{vmatrix} = \frac{1}{R \operatorname{sen}\theta}\left[\frac{\partial(\operatorname{sen}\theta F_\phi)}{\partial \theta} - \frac{\partial F_\theta}{\partial \phi}\right] \hat{a}_n +$$

$$\frac{1}{R}\left[\frac{1}{\operatorname{sen}\theta} \frac{\partial F_n}{\partial \phi} - \frac{\partial (R F_\phi)}{\partial R}\right] a_\theta + \left[\frac{\partial (R F_\theta)}{\partial R} - \frac{\partial F_n}{\partial \theta_x}\right] a_\phi$$

$$\begin{cases} F_\phi = 0 \\ F_\theta = 0 \\ F_R = R \cos\theta \end{cases}, \Rightarrow -\frac{\partial F_R}{\partial \theta} = \frac{\partial (R \operatorname{sen}\theta)}{\partial \theta} = \boxed{R \cos\theta}$$

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